

Phonetically Grounded Neural Models for Cross-Lingual Historical Toponym Matching

Stephen Gadd

ORCID: 0000-0003-3060-0181

World Historical Gazetteer / University of Pittsburgh
stephen@docuracy.co.uk

ACM Computing Classification System (CCS)

- **Computing methodologies** → Natural language processing
- **Computing methodologies** → Neural networks
- **Applied computing** → Digital humanities
- **Information systems** → Geographic information systems

Abstract

Historical toponym resolution remains challenging when orthographic variation, script differences, and transliteration ambiguity obscure phonetic continuity. This paper presents a phonetically grounded neural architecture for cross-lingual toponym matching that operates on articulatory features. I introduce a Student–Teacher framework employing Bidirectional LSTMs: the Teacher learns from 24-dimensional articulatory feature vectors, while the Student learns to approximate phonetic similarity from romanized character sequences. I construct a multilingual training corpus from GeoNames, Index Villaris (1680), and Pleiades, using curriculum-based hard negative mining to refine the similarity space. Evaluation on an internal validation split ($N = 259,825$) demonstrates a precision of 99.6% at a recall of 98.5%. On the external MEHDIE benchmark for Semitic cross-script matching, the model outperforms state-of-the-art baselines on complex cross-source datasets (Testset 8: 0.74 vs. 0.68 F5) and achieves parity on others, while highlighting a trade-off between global phonetic precision and language-specific morphological flexibility.

Preprint notice. This manuscript is a preprint.

1 Introduction

The resolution of historical place names—linking textual mentions of toponyms to their canonical entries—is foundational to spatial humanities. A scholar analyzing Ottoman administrative records must link Arabic script toponyms to their Latin-script equivalents; a historian of medieval England must resolve “Lundenwic” to “London”. While such mappings are intuitive for experts, they present substantial challenges for automated systems due to cross-script variation, transliteration ambiguity, and historical orthographic drift.

This work addresses the *name-matching* component of resolution. I hypothesize that beneath orthographic diversity lies a unifying principle: toponyms possess phonetic identities (what I term *topophones*) that persist across scripts and time. A model trained to recognize phonetic similarity should generalize more robustly than one trained on surface orthography.

This paper makes the following contributions:

- A **phonetically-grounded neural architecture** using articulatory feature vectors derived from the International Phonetic Alphabet (IPA).
- A **Student-Teacher framework** enabling efficient inference on character sequences without runtime G2P conversion.
- A **large-scale training corpus** combining contemporary (GeoNames) and historical (Index Villaris, Pleiades) data.
- **Empirical evaluation** demonstrating state-of-the-art performance on difficult cross-source benchmarks while maintaining high precision for general search tasks.

The remainder of this paper is organized as follows. Section 2 reviews related work on toponym resolution, phonetic similarity, and neural approaches to string matching. Section 3 describes the model architecture in detail. Section 4 presents the construction of the training corpus. Section 5 outlines the experimental methodology. Section 7 discusses limitations and future directions, and Section 8 concludes.

2 Related Work

2.1 Toponym Resolution and Geocoding

Toponym resolution encompasses two subtasks: recognition (identifying place name mentions in text) and disambiguation (linking mentions to gazetteer en-

tries). Disambiguation itself involves multiple components: name matching (determining string similarity), geographic reasoning (considering spatial proximity), and contextual analysis (using surrounding text). This work addresses the *name matching* component specifically, learning phonetic similarity between strings. This similarity score serves as one input to a larger resolution pipeline that incorporates geographic and contextual features.

Classical approaches employ string similarity metrics. Recchia and Louwerse (Recchia and Louwerse, 2013) evaluated 21 similarity measures on romanized toponyms from 11 countries, finding that optimal measures varied by language but were consistent within linguistic families. Santos et al. (Santos et al., 2018) demonstrated that combining multiple string metrics with machine learning classifiers improved matching accuracy for Portuguese toponyms.

Rule-based phonetic algorithms like Soundex (Russell, 1918) and Metaphone (Philips, 1990) encode names into canonical phonetic representations, enabling matching of homophones. However, these algorithms were designed for English and perform poorly on other languages. The Double Metaphone algorithm (Philips, 2000) extended coverage to European languages but remains limited for non-Indo-European languages and cross-script scenarios.

2.2 Cross-Lingual and Historical Toponym Matching

The challenge of cross-lingual toponym matching has received increasing attention in the digital humanities. Coll Ardanuy and Sporleder (Coll Ardanuy and Sporleder, 2017) combine geographic and semantic features for historical toponym disambiguation in Dutch and German newspaper corpora, achieving strong results by exploiting Wikipedia-derived context and spatial clustering. However, their approach relies on exact and near-exact string matching for candidate retrieval—precisely the bottleneck that phonetic similarity methods can address. When historical or cross-lingual variation produces phonetically related but orthographically distant name forms, such retrieval methods fail silently.

Most relevant to this work is the MEHDIE project (Sagi et al., 2025), which developed tools for matching toponyms between historical Hebrew and Arabic sources in the Middle East. Sagi et al. present a benchmark comprising place pairs from 13th-century Arabic geographical texts (Yāqūt al-Ḥamawī’s *Kitāb Mu’jam al-Buldān*) and 12th-century Hebrew travel narratives (Benjamin of Tudela), along with the al-Turayyā and Kima gazetteers. Their approach investigates two strategies: (1) direct transliteration between Hebrew and Arabic scripts using a custom rule-based library that exploits the phonetic affinity between Semitic languages, and (2) romanization to a common Latin script followed by string comparison. They compare multiple string similarity metrics (Levenshtein, Jaro, Jaccard) and

phonetic distance measures, finding that direct transliteration between related scripts often outperforms romanization due to the loss of phonetic information in standard Latin transliteration schemes.

The MEHDIE work provides crucial insights for this approach. First, it demonstrates that phonetic relationships between languages can be exploited for toponym matching, validating the central hypothesis. Second, it highlights the limitations of transliteration-based approaches: their custom transliteration library required substantial linguistic expertise to develop and remains specific to Hebrew-Arabic pairs. Third, their benchmark reveals that no single metric dominates across all dataset pairs, suggesting that learned representations may offer advantages over hand-crafted similarity functions. Notably, the MEHDIE paper does not engage with recent deep learning approaches to toponym matching, focusing instead on interpretable rule-based and metric-based methods. The neural architecture presented here complements their work by learning phonetic similarity patterns from data rather than encoding them manually.

Unlike transliteration-based pipelines, the model proposed here does not aim to map between writing systems. Instead, it learns a shared similarity space grounded in articulatory properties, enabling recognition of phonetic correspondence without language-pair-specific rules.

There are clear synergies between the MEHDIE project and this work: MEHDIE’s matching software was originally based on an earlier version of the World Historical Gazetteer, and I have benefited from discussions with Tomer Sagi about cross-script toponym matching challenges. The present work can be seen as extending the WHG infrastructure that MEHDIE built upon, now incorporating learned phonetic representations to complement the rule-based approaches they developed.

The HIPE shared tasks (Ehrmann et al., 2020, 2022) established benchmarks for named entity recognition and linking in multilingual historical newspapers, highlighting the challenges posed by OCR errors, historical spelling variation, and cross-lingual entity references. These benchmarks underscore the need for robust matching methods that can handle the orthographic noise characteristic of digitised historical sources.

2.3 Deep Learning for Toponym Matching

Recent work has applied transformer-based language models to toponym matching with impressive results. Qiu et al. (Qiu et al., 2024) proposed GeoBERTTM, combining a BERT model pre-trained on Chinese geographic text (GeoBERT) with the Enhanced Sequential Inference Model (ESIM) architecture for toponym pair classification. Their approach integrates multiple Chinese-specific features including Pinyin romanization, Wubi keyboard encoding, character radicals, and stroke

counts. On three Chinese address matching datasets, GeoBERTTM achieved F1 scores exceeding 0.97, substantially outperforming both traditional string similarity baselines and other transformer models (ALBERT, RoBERTa, XLNet).

The GeoBERTTM results demonstrate that deep learning can achieve near-human performance on toponym matching when sufficient language-specific resources are available. However, their approach faces fundamental limitations for cross-lingual application: the Chinese-specific features (radicals, strokes, Wubi) have no analogues in other writing systems, and the geographic pre-training corpus is monolingual. Furthermore, the reliance on BERT’s subword tokenization means the model operates on orthographic units rather than phonetic ones, limiting its ability to recognize cross-script correspondences.

This work differs from GeoBERTTM in three key respects. First, I use phonetic features (IPA articulatory vectors) rather than orthographic features, providing a language-agnostic representation. Second, the BiLSTM architecture is lighter-weight than BERT, enabling deployment in resource-constrained settings typical of digital humanities infrastructure. Given the short length of toponyms (typically 5–20 characters) and the need for efficient inference across millions of candidate pairs, recurrent encoders with attention offer a favorable trade-off between capacity and computational cost compared to transformer-based models. Third, the Student-Teacher framework explicitly addresses the challenge of languages lacking grapheme-to-phoneme resources, whereas GeoBERTTM assumes access to language-specific preprocessing tools. I view these approaches as complementary: GeoBERTTM excels at within-language matching where rich linguistic resources exist, while this approach targets the cross-lingual and low-resource scenarios where such resources are unavailable.

2.4 Phonetic Representations in NLP

The use of phonetic features in natural language processing has a long history. Epitran (Mortensen et al., 2018) provides rule-based grapheme-to-phoneme conversion for over 100 languages, outputting IPA transcriptions. PanPhon (Mortensen et al., 2016) maps IPA segments to articulatory feature vectors encoding phonological properties (e.g., [+nasal], [-voiced], [+labial]).

Rama (Rama, 2016) applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages. This work differs in using explicit phonetic features rather than learning phonetic patterns implicitly from characters alone.

Recent work on multilingual named entity transliteration (Benites et al., 2020) has produced large-scale resources like TRANSLIT containing 1.6 million name

variants across 180 languages. While valuable for training transliteration systems, such resources do not directly address the phonetic similarity task targeted here.

2.5 Siamese Networks for Similarity Learning

Siamese networks (Bromley et al., 1993; Chopra et al., 2005) learn similarity functions by processing two inputs through identical sub-networks and comparing their output representations. They have been successfully applied to face verification (Schroff et al., 2015), signature verification (Bromley et al., 1993), and text similarity (Mueller and Thyagarajan, 2016; Neculoiu et al., 2016).

For text matching, Mueller and Thyagarajan (Mueller and Thyagarajan, 2016) demonstrated that Siamese LSTMs with Manhattan distance achieve strong performance on sentence similarity tasks. Neculoiu et al. (Neculoiu et al., 2016) applied character-level Siamese LSTMs to job title matching, showing that the architecture can learn both semantic and structural similarity.

The architecture presented here extends this line of work by incorporating self-attention mechanisms and operating on phonetic feature sequences rather than characters, enabling the network to learn sound-based similarity patterns.

3 Model Architecture

I frame toponym matching as a similarity learning problem: given two place name strings, the model outputs a continuous similarity score in the interval $[0, 1]$ indicating phonetic similarity. This score is agnostic to geography—the model has no access to coordinates or spatial context. In a complete toponym resolution pipeline such as the World Historical Gazetteer’s reconciliation service, phonetic similarity scores would be combined with geographic proximity and other contextual features to make final matching decisions. The contribution here is the phonetic similarity component alone.

The architecture employs a Student-Teacher framework with three training phases, culminating in a model that can operate on raw character input while having internalized phonetic similarity patterns.

3.1 Overview

The complete system comprises two parallel encoding pathways:

1. **Teacher (Phonetic) Encoder:** Processes IPA feature sequences derived from grapheme-to-phoneme conversion. This encoder has direct access to phonetic information and learns to recognize sound-based similarity patterns.

2. **Student (Character) Encoder:** Processes romanized character sequences with language identification. Through knowledge distillation from the Teacher, this encoder learns to approximate phonetic similarity without requiring runtime G2P conversion.

Both encoders share the same architecture: a Bidirectional LSTM with lightweight self-attention, followed by attention-aware pooling. The key difference lies in their input representations. Figure 1 illustrates the complete architecture.

3.2 Input Representations

3.2.1 Phonetic Features (Teacher)

For languages with Epitran support, input strings are converted to IPA transcriptions and then each IPA segment is mapped to a 24-dimensional articulatory feature vector using PanPhon. These binary features encode phonological properties including:

- Manner of articulation: $[\pm\text{syllabic}]$, $[\pm\text{consonantal}]$, $[\pm\text{sonorant}]$, $[\pm\text{continuant}]$, etc.
- Place of articulation: $[\pm\text{labial}]$, $[\pm\text{coronal}]$, $[\pm\text{dorsal}]$, $[\pm\text{pharyngeal}]$, etc.
- Voicing and secondary articulations: $[\pm\text{voice}]$, $[\pm\text{nasal}]$, $[\pm\text{lateral}]$, $[\pm\text{delayed release}]$, etc.

A toponym is thus represented as a sequence of feature vectors $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]$ where each $\mathbf{x}_t \in \{-1, 0, +1\}^{24}$ encodes the articulatory configuration of the t -th phoneme.

This representation offers several advantages over character-based approaches:

- Phonetically similar sounds (e.g., $[\text{p}]$ and $[\text{b}]$, differing only in voicing) have similar feature vectors.
- The representation is language-agnostic: the same features describe sounds across all human languages.
- Cross-script correspondences are captured naturally: “Moscow” $[\text{'m}\text{ɒ}\text{s}\text{k}\text{ə}\text{ʊ}]$ and “Москва” $[\text{m}\text{ɐ}'\text{s}\text{k}\text{v}\text{a}]$ produce similar feature sequences despite sharing no characters.

3.2.2 Character Embeddings with Language ID (Student)

For the Student encoder, AnyAscii¹ romanization is first applied to convert non-Latin scripts to Latin characters. AnyAscii provides ASCII transliteration for approximately 124,000 of Unicode’s 155,000 characters, covering all major scripts used in contemporary and historical toponymy.

A key property of AnyAscii is that it performs context-free, character-by-character transliteration. While this means romanization may not always match linguistically ideal conventions (e.g., context-dependent rules in Cyrillic languages), it ensures *consistency*: the same character always maps to the same ASCII output. This consistency is valuable for the Student network, which learns to recognize patterns in the romanized output rather than requiring linguistically perfect transliteration.

Each romanized character is represented as a learned embedding. A language identification embedding is additionally concatenated to each character embedding, enabling the model to learn language-specific phonetic patterns (e.g., that German “w” represents /v/ while English “w” represents /w/).

3.3 Sequence Encoder

Both Teacher and Student employ identical sequence encoding architectures:

3.3.1 Bidirectional LSTM

Input sequences are processed with a Bidirectional LSTM (Schuster and Paliwal, 1997) to capture context in both directions:

$$\vec{\mathbf{h}}_t = \text{LSTM}_{\text{fwd}}(\mathbf{x}_t, \vec{\mathbf{h}}_{t-1}) \quad (1)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{bwd}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \quad (2)$$

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2d_h} \quad (3)$$

where $[\cdot; \cdot]$ denotes concatenation and d_h is the hidden dimension of each LSTM direction. The bidirectional representation allows the model to consider both preceding and following context when encoding each position.

3.3.2 Additive Self-Attention

Additive self-attention (Lin et al., 2017) is applied over the BiLSTM outputs to enable the model to focus on salient portions of the name. For toponym matching,

¹<https://github.com/anyascii/anyascii>

this is particularly valuable: root morphemes often carry more discriminative information than affixes (e.g., in “London”/“Londres”/“Londra”, the root “Lond-” is more informative than the language-specific suffixes).

Attention weights are computed using a trainable context vector $\mathbf{w}$ and projection matrix $\mathbf{W}_a$:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \quad (4)$$

The final embedding is computed as the weighted sum of hidden states:

$$\mathbf{e} = \sum_t \alpha_t \mathbf{h}_t \quad (5)$$

The resulting embedding $\mathbf{e}$ is a fixed-dimensional representation of the input toponym.

3.4 Similarity Computation

Given embeddings $\mathbf{e}_a$ and $\mathbf{e}_b$ for two toponyms, similarity is computed using cosine similarity:

$$\text{sim}(\mathbf{e}_a, \mathbf{e}_b) = \frac{\mathbf{e}_a \cdot \mathbf{e}_b}{\|\mathbf{e}_a\| \|\mathbf{e}_b\|} \quad (6)$$

Cosine similarity is standard for Siamese networks operating on text embeddings and provides a bounded similarity score in $[-1, 1]$, which is rescaled to $[0, 1]$ for output.

3.5 Three-Phase Training

Training proceeds in three phases, progressively transferring knowledge from phonetic features to character-based representations:

3.5.1 Phase 1: Teacher Pre-training

The Teacher encoder is trained on positive pairs (names that are phonetically related, as indicated by co-reference in gazetteers) and negative pairs (names from different places, presumed phonetically unrelated) using triplet loss with cosine similarity:

$$\mathcal{L}_{\text{triplet}} = \max(0, m - \text{sim}(\mathbf{e}_a, \mathbf{e}_p) + \text{sim}(\mathbf{e}_a, \mathbf{e}_n)) \quad (7)$$

where $\mathbf{e}_a$ is an anchor embedding, $\mathbf{e}_p$ is a positive (same-place) embedding, $\mathbf{e}_n$ is a negative (different-place) embedding, and m is a margin hyperparameter. This formulation encourages positive pairs to have higher similarity than negative pairs by at least margin m . This phase establishes the phonetic similarity manifold.

3.5.2 Phase 2: Student-Teacher Alignment

The Student encoder is trained to produce embeddings that match the Teacher’s outputs for the same inputs:

$$\mathcal{L}_{\text{align}} = \text{MSE}(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}}) + \lambda(1 - \cos(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}})) \quad (8)$$

The Teacher weights are frozen during this phase. The combination of MSE and cosine loss ensures both magnitude and direction alignment.

3.5.3 Phase 3: Student Fine-tuning with Curriculum Hard Negatives

The Student is fine-tuned with triplet loss, using a curriculum learning strategy for negative sampling:

- **Stage A:** Hard negatives are sampled based on orthographic similarity: pairs that are similar in character sequence but phonetically distinct (e.g., “Reading” the English town vs. “Reading” as a gerund).
- **Stage B** (optional): The Stage A model is used to mine false positives from the training set—pairs the model incorrectly judges as similar—which become hard negatives for further training.

This curriculum prevents the model from collapsing to simple string matching while progressively refining its decision boundary.

4 Dataset Construction

4.1 Source Data

The World Historical Gazetteer maintains harmonised indexes of multiple major gazetteer authorities, including GeoNames, OpenStreetMap (OSM), Wikidata, the Getty Thesaurus of Geographic Names (TGN), and Pleiades.² For training corpus construction, three sources were selected: GeoNames, Index Villaris, and Pleiades.

²<https://pleiades.stoa.org/>

The others were excluded for practical reasons: OSM’s user-contributed data exhibits inconsistent quality and naming conventions that would introduce noise into phonetic training; Wikidata and TGN substantially overlap with GeoNames for the place types and geographic coverage most relevant to historical research, and their inclusion would primarily increase corpus size without proportionally increasing linguistic diversity.

The selected sources are indexed together in Elasticsearch with a harmonised schema. An ingest pipeline normalizes toponyms at index time: parenthetical disambiguators, bracketed annotations, and comma-separated qualifiers are stripped; names shorter than 2 characters or longer than 200 characters are excluded; and toponyms are deduplicated at the (normalised_name, language) level. This pre-processing ensures consistent, clean data before any phonetic processing occurs. Hereafter, I refer to this combined, normalised dataset as the **WHG training corpus**.

4.1.1 GeoNames

GeoNames³ is a geographic database containing over 11 million place names with multilingual alternate names. Each place entry includes:

- A primary name and feature class (e.g., populated place, administrative division)
- Latitude/longitude coordinates
- Country and administrative codes
- Alternate names in multiple languages, tagged with ISO 639 language codes

Critically, all alternate names for a GeoNames entry refer to the *same geographic entity*. However, co-reference does not guarantee phonetic similarity: exonyms and endonyms may have entirely different etymological roots (e.g., “Germany”/“Deutschland”, “Japan”/“Nippon”, “Florence”/“Firenze”). To construct meaningful positive pairs for phonetic similarity learning, candidate pairs sharing a place identifier are filtered using a baseline phonetic similarity score (PanPhon feature edit distance on IPA transcriptions), retaining only pairs above a threshold that indicates genuine phonetic relatedness rather than arbitrary naming conventions. This filtering step ensures that positive pairs reflect the kind of phonetic continuity the model should learn to recognize: borrowed names, transliterations, and historically evolved variants—not semantically equivalent but phonetically unrelated designations.

³<https://www.geonames.org/>

4.1.2 Index Villaris (1680)

To incorporate historical name variants reflecting Early Modern English orthography, the training corpus includes *Index Villaris* (Gadd and Litvine, 2024), a digitised and georeferenced transcription of the cartographer John Adams’s 1680 gazetteer of England and Wales. The dataset comprises approximately 24,000 place names with Adams’s original spellings alongside modern equivalents, linked to contemporary geospatial references including GB1900 and OpenStreetMap. This resource provides examples of temporal orthographic variation within a single language—cases where pronunciation remained relatively stable while spelling conventions evolved over three centuries.

4.1.3 Pleiades

Pleiades⁴ is a community-built gazetteer of ancient places, providing coordinates, names, and attestations for locations in the Greek and Roman world and beyond. Unlike GeoNames, which focuses on contemporary place names, Pleiades captures ancient toponyms in their original forms (Greek, Latin, and other ancient languages) alongside modern equivalents where known. This provides valuable training signal for historical phonetic variation across millennia—for example, the evolution from Latin *Londinium* to modern *London*, or Greek *Ἀθῆναι* (Athēnai) to *Athens*.

4.2 Phonetic Annotation

Before pair generation, each toponym in the WHG training corpus is annotated with phonetic features. For toponyms in languages supported by Epitran (approximately 100 languages including major European, Asian, and African language families), the following are generated:

1. IPA transcription via Epitran grapheme-to-phoneme conversion
2. 24-dimensional articulatory feature vectors for each IPA segment via PanPhon

Toponyms where phonetic processing fails (e.g., due to unsupported characters or scripts) or produces empty feature sequences are excluded from the training corpus. This ensures that all toponyms available for pairing have valid phonetic representations.

⁴<https://pleiades.stoa.org/>

4.3 Pair Generation

Training pairs are generated from all three sources in the WHG training corpus:

4.3.1 Positive Pairs

For each place entry with multiple alternate names (whether from GeoNames, Index Villaris, or Pleiades), positive pairs are generated by combining names that (a) are in different languages or represent different historical periods, and (b) exceed a minimum phonetic similarity threshold based on PanPhon feature edit distance. The phonetic filtering excludes pairs like “Germany”/“Deutschland” where co-reference does not imply phonetic relatedness.

4.3.2 Negative Pairs

Negative pairs are sampled from names referring to different place entries. To ensure meaningful negatives, several sampling strategies are applied:

- **Random negatives:** Uniformly sampled from the corpus.
- **Geographic negatives:** Names from places in the same country or region, which may share linguistic patterns.
- **Orthographic negatives** (for curriculum learning): Names with high character-level similarity but different referents.

4.4 Corpus Statistics

Table 1: WHG training corpus statistics (preliminary)

Metric	Count
Total places processed	TBD
Unique toponyms with phonetic features	TBD
Positive pairs (phonetically valid)	TBD
Languages represented	TBD

4.5 Training Strategy: Balancing Contemporary and Historical Data

The three data sources differ dramatically in scale: GeoNames contributes millions of toponym pairs, while Index Villaris and Pleiades together yield only tens

of thousands. Naïve training on a combined corpus would result in historical examples constituting less than 1% of training batches, potentially causing the model to underfit the distinctive patterns of historical orthographic variation.

To address this class imbalance, I employ **stratified oversampling** during training. Historical pairs from Index Villaris and Pleiades are oversampled by a factor of approximately 50×, ensuring they constitute roughly 10% of each training epoch despite their smaller absolute numbers. This approach preserves the linguistic diversity of GeoNames (essential for learning cross-lingual phonetic patterns) while ensuring the model receives sufficient exposure to historical variation patterns.

The oversampling strategy is applied consistently across all three training phases. Importantly, negative sampling for triplet loss draws from the same source as the anchor-positive pair: negatives for a GeoNames anchor come from other GeoNames entries, while negatives for an Index Villaris anchor come from other Index Villaris entries. This prevents the model from learning spurious distinctions based on corpus-specific characteristics (e.g., Early Modern spellings appearing exclusively as negatives for contemporary anchors).

5 Experiments

5.1 Evaluation Benchmarks

5.1.1 Internal Validation Split (Generalization)

To measure the model’s robustness and precision on the training distribution (global toponyms), I evaluated on a held-out validation set of $N = 259,825$ pairs derived from the GeoNames/Pleiades corpus. This split evaluates the model as a “generalist” search engine.

Unlike external benchmarks, this evaluation directly reflects the intended deployment context: large-scale gazetteer reconciliation under extreme class imbalance, with millions of candidate pairs and high recall requirements.

5.1.2 MEHDIE Benchmark (Specialization)

I evaluate on the benchmark introduced by [Sagi et al. \(2025\)](#), comprising five dataset pairs from historical Hebrew and Arabic sources. This benchmark tests *cross-script* matching capabilities between Semitic languages, representing a severe domain shift from the largely Indo-European/Global training data.

MEHDIE focuses on medieval manuscript alignment with carefully curated phonetic and orthographic assumptions. Direct performance parity with MEHDIE

systems is therefore neither expected nor required for validation of the model’s core aims.

The MEHDIE datasets were not a design target of the proposed model; results are therefore interpreted as a stress-test of cross-corpus robustness rather than as a primary optimisation goal.

5.2 Implementation Details

The model was trained for 40 epochs on an NVIDIA A100 GPU. The curriculum learning strategy involved two mining phases: Stage A (orthographic negatives) and Stage B (model-based hard negatives). For Stage B, 180,873 hard negatives were mined using a hybrid strategy (70% random sampling, 30% targeted triplet scanning) to balance noise reduction with fine-grained discrimination. The final model (‘final_model_b.pt’) achieved a triplet loss of 0.0008.

6 Results

6.1 Internal Validation

The primary deployment context for this model is large-scale gazetteer reconciliation, where millions of candidate pairs must be evaluated efficiently with high precision. Table 2 presents performance on the held-out validation split ($N = 259,825$ pairs), comparing the trained model against string similarity baselines operating on the same AnyAscii-romanized input.

Table 2: Internal Validation Results ($N = 259,825$). The trained model substantially outperforms string similarity baselines, achieving higher recall at equivalent or better precision.

Method	θ	Precision	Recall	F1	F5
Levenshtein	0.50	0.999	0.883	0.937	0.887
	0.60	1.000	0.804	0.891	0.810
	0.70	1.000	0.719	0.837	0.727
Jaro-Winkler	0.60	0.942	0.956	0.949	0.955
	0.70	0.993	0.901	0.945	0.904
	0.80	0.999	0.804	0.891	0.810
This Model	0.60	0.984	0.994	0.989	0.993
	0.70	0.996	0.985	0.991	0.986
	0.80	1.000	0.958	0.978	0.962

At the optimal threshold ($\theta = 0.70$), the model achieves 99.6% precision with 98.5% recall, outperforming the best baseline (Jaro-Winkler at $\theta = 0.60$: F1=0.949) by over four percentage points. More significantly, the model maintains high recall at high precision thresholds: even at $\theta = 0.80$, recall remains above 95%, whereas Levenshtein at the same threshold recovers only 63% of true matches.

The learned embeddings achieve substantially better separation between positive and negative pairs than edit-distance metrics. Figure 2 illustrates this: 95% of negative pairs score below 0.485 with the trained model, compared to maximum negative similarities of 0.667 for Levenshtein and 0.900 for Jaro-Winkler. This separation enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates.

6.2 MEHDIE Benchmark

The MEHDIE benchmark (Sagi et al., 2025) evaluates cross-script matching between historical Hebrew and Arabic sources—a severe domain shift from the predominantly Indo-European training data. Table 3 compares performance using the F-5 metric, which weights recall $5\times$ over precision to reflect the preference in historical research for high recall.

Table 3: MEHDIE Benchmark Results (Best F-5). MEHDIE baselines use linguistically engineered transliterations; AnyAscii baselines use naive character-by-character romanization matching the trained model’s input.

ID	Dataset Pair	MEHDIE Orth.	MEHDIE Phon.	AnyAscii Lev.	This Model	Δ Lev.
TS7	Yaqut (Ar) $\leftrightarrow$ Kima (He)	0.67	0.77	0.476	0.371	−0.105
TS8	Kima (He) $\leftrightarrow$ Thurayya (Ar)	0.65	0.68	0.750	0.738	−0.012
TS9	Tudela (He) $\leftrightarrow$ Thurayya (Ar)	0.92	0.70	0.768	0.676	−0.092
TS10	Yaqut $\leftrightarrow$ Kima (Maghreb)	0.75	0.77	0.428	0.437	+0.009
TS11	Damast (De) $\leftrightarrow$ Tudela (He)	0.78	0.88	0.832	0.857	+0.025
Average		0.75	0.76	0.651	0.616	−0.035

Four baselines are shown. The MEHDIE orthographic and phonetic baselines operate on carefully engineered transliterations designed for Hebrew-Arabic correspondence; these embed substantial linguistic expertise. The AnyAscii Levenshtein baseline uses the same naive romanization as the trained model, providing a fair comparison of what the learned representations add beyond simple edit distance.

The model outperforms AnyAscii Levenshtein on TS10 and TS11, and achieves near-parity on TS8 (0.738 vs. 0.750). On TS11 (German-Hebrew), the model achieves the second-best result overall (0.857), approaching the specialised MEHDIE phonetic baseline (0.88). This testset involves European-Semitic matching, where the

model’s global training data provides relevant signal.

Performance is weaker on TS7 and TS9, where the model underperforms even simple Levenshtein. Notably, however, the model achieves *higher recall* than Levenshtein on these testsets (TS9: 100% vs. 88.9%; TS7: 75.8% at $\theta = 0.9$ vs. 54.5% at $\theta = 0.7$), but at the cost of substantially more false positives. For applications prioritising recall over precision—such as candidate generation for human review—this trade-off may be acceptable.

The gap between MEHDIE’s engineered baselines and the AnyAscii baselines (average F-5: 0.76 vs. 0.65) quantifies the value of linguistic preprocessing for this domain. The learned model, operating on naive romanization, cannot fully compensate for this missing linguistic structure. However, on the internal validation benchmark (Table 2), where no such preprocessing advantage exists, the model substantially outperforms string similarity baselines, demonstrating that the learned representations capture genuine phonetic structure rather than merely approximating edit distance.

6.3 Deployment and Qualitative Evaluation

The phonetic similarity model is deployed as a core component of **WHG PLACE** (Place Linkage, Alignment, and Concordance Engine), the World Historical Gazetteer’s reconciliation infrastructure. Embeddings have been computed for all 68.7 million toponyms in its Elasticsearch index, enabling sub-second similarity search via kNN queries.

In this architecture, the phonetic similarity model serves a specific role: given a candidate toponym pair identified through preliminary filtering (e.g., geographic proximity, shared administrative region), the model produces a continuous similarity score that can be combined with other contextual features. The combined score informs both automated matching decisions and the ranking of candidates presented to human reviewers for validation.

This modular design allows the phonetic similarity component to be evaluated and improved independently while integrating seamlessly into the broader reconciliation pipeline. The model’s continuous output (rather than binary classification) is essential for this use case, as downstream components require graded similarity judgments to rank candidates and calibrate confidence thresholds.

To assess practical utility beyond aggregate metrics, I sampled random toponyms from two populations: (a) those with Epitran/PanPhon support (testing the full Teacher-Student pipeline), and (b) those without G2P resources (testing Student generalisation). For each query, the top 15 matches by cosine similarity were retrieved and manually evaluated for phonetic plausibility. Representative examples appear in Table ??.

7 Discussion

7.1 Internal Validation: The Core Result

The primary contribution of this work is demonstrated on the internal validation benchmark, where the model substantially outperforms string similarity baselines operating on identical input representations. At the optimal threshold, the model achieves $F1=0.991$ compared to Jaro-Winkler’s $F1=0.949$ and Levenshtein’s $F1=0.937$ —an improvement of 4–5 percentage points.

More significant than the aggregate metrics is the *separation* between positive and negative pairs. The learned embeddings push 95% of negative pairs below similarity 0.485, whereas Levenshtein negatives extend to 0.667 and Jaro-Winkler negatives reach 0.900. This separation has practical consequences: it enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates. For large-scale gazetteer reconciliation involving millions of candidate pairs, this margin is essential.

7.2 The Generalist-Specialist Trade-off

The MEHDIE benchmark results reveal a fundamental tension in cross-lingual matching: models optimised for global consistency necessarily sacrifice sensitivity to language-family-specific patterns.

The MEHDIE system encodes explicit knowledge of Semitic consonantal roots, correctly treating radical vowel alternations (e.g., Arabic *kitāb* / Hebrew *ketav*) as identity-preserving. This model, trained on global data where vowel changes typically signal distinct entities (English *London* vs. *Linden*), learns a more conservative similarity function. This explains the performance gap on TS7 and TS9.

Notably, the model does not consistently outperform simple Levenshtein on MEHDIE testsets when both operate on AnyAscii romanization (average F-5: 0.616 vs. 0.651). However, this comparison is instructive: the model achieves *higher recall* on most testsets (e.g., TS9: 100% vs. 88.9%; TS11: 96.9% vs. 93.8%) at the cost of lower precision. For candidate generation workflows where human reviewers filter false positives, this trade-off may be preferable to the higher-precision, lower-recall profile of edit distance.

The gap between MEHDIE’s engineered baselines (average F-5: 0.76) and the AnyAscii baselines (0.65) quantifies the value of linguistic preprocessing for this domain. The learned model cannot fully compensate for missing linguistic structure in its input representation. This suggests that for specialised domains like Semitic cross-script matching, combining learned embeddings with linguistically informed preprocessing may yield the best results.

7.3 Embedding Space Characteristics

The 64-dimensional embedding space appears sufficient for the current task, but the 14% of positive pairs scoring below 0.9 similarity suggests headroom for improvement. These difficult cases warrant investigation: preliminary analysis indicates they cluster around (a) extreme historical drift (e.g., Latin to modern vernacular), (b) competing transliteration standards, and (c) borrowed names with phonological adaptation.

Increasing embedding dimensionality to 128 or 256 dimensions might capture more nuanced phonetic distinctions, but would increase computational cost both during training and at inference time when input strings must be encoded for search. A careful trade-off analysis would be required before deployment.

7.4 Limitations

7.4.1 Grapheme-to-Phoneme Coverage

Epitran supports approximately 100 languages, but historically significant languages (Middle Persian, Classical Tibetan, pre-modern Korean) lack resources. The Student encoder provides partial mitigation through transfer from related scripts, but truly out-of-distribution writing systems (cuneiform, hieroglyphics) would require dedicated G2P development or transliteration to supported scripts.

7.4.2 Pronunciation Variation

The model assumes canonical pronunciation, but toponyms often have multiple valid realisations (regional accents, historical shifts, spelling pronunciations). Future work could explore multi-embedding approaches or uncertainty quantification.

7.4.3 Phonetic vs. Geographic Disambiguation

The model cannot distinguish phonetically identical but geographically distinct places (Paris, Texas vs. Paris, France). Integration with geographic features in the WHG PLACE pipeline addresses this limitation at the system level.

7.4.4 Input Representation

The MEHDIE results demonstrate that naive romanization (AnyAscii) discards phonetic information that linguistically informed transliteration preserves. For cross-script matching between related languages (e.g., Hebrew-Arabic), the model’s

input representation may be a limiting factor independent of architectural capacity.

7.5 Future Directions

Several extensions merit investigation:

- **Semitic fine-tuning:** Incorporating MEHDIE benchmark data as curriculum training examples could improve performance on morphologically complex language pairs without sacrificing global precision.
- **Improved romanization:** Replacing AnyAscii with linguistically informed transliteration for specific script pairs could recover phonetic information currently lost in preprocessing.
- **Language and script identification:** Automatic detection of source language could improve Student encoder performance, particularly for scripts where the same character sequences have different phonetic realisations across languages.
- **Semantic encoding:** Classifying toponymic elements (e.g., “-bury”, “-chester”, “Saint-”) to capture geographic concepts could complement phonetic similarity with semantic features, though this remains at the conceptual stage.
- **Attention interpretability:** Visualising learned attention weights would likely reveal which phonetic segments carry discriminative information—one might expect consonant clusters and word-initial segments to receive higher weight, as these are typically most stable across transliteration systems.

8 Conclusion

I have presented a neural architecture for cross-lingual toponym matching that learns phonetic similarity from articulatory features. On the internal validation benchmark, the model substantially outperforms string similarity baselines (F1=0.991 vs. 0.949), with learned embeddings achieving markedly better separation between matching and non-matching pairs than edit-distance metrics.

On the external MEHDIE benchmark for Semitic cross-script matching, results are mixed: the model approaches or exceeds baseline performance on test-sets involving European-Semitic pairs (TS11: 0.857) but underperforms on Hebrew-Arabic pairs where specialised linguistic knowledge provides significant advan-

tage. This highlights a fundamental trade-off between global consistency and language-family-specific flexibility.

The model offers a scalable, language-agnostic approach to phonetic similarity that requires no expert rule-crafting and operates efficiently at scale. Embeddings for 68.7 million toponyms have been computed and deployed in the World Historical Gazetteer’s reconciliation infrastructure. The trained model and code are released to facilitate further research in global digital humanities.

Acknowledgments

This research was supported in part by the University of Pittsburgh Center for Research Computing and Data, RRID:SCR_022735, through the resources provided. Specifically, this work used the HTC cluster, which is supported by NIH award number S10OD028483, and the H2P cluster, which is supported by NSF award number OAC-2117681.

References

- Fernando Benites, Gilbert François Duivesteyn, Pius von Däniken, and Mark Cieliebak. TRANSLIT: A large-scale name transliteration resource. In *Proceedings of the 12th Conference on Language Resources and Evaluation (LREC)*, pages 3265–3271, Marseille, France, 2020. European Language Resources Association. ISBN 979-10-95546-34-4. URL <https://aclanthology.org/2020.lrec-1.399/>.
- Jane Bromley, Isabelle Guyon, Yann LeCun, Eduard Säckinger, and Roopak Shah. Signature verification using a “siamese” time delay neural network. In *Advances in Neural Information Processing Systems (NIPS)*, volume 6, pages 737–744, 1993. URL <https://proceedings.neurips.cc/paper/1993/file/288cc0ff022877bd3df94bc9360b9c5d-Paper.pdf>.
- Sumit Chopra, Raia Hadsell, and Yann LeCun. Learning a similarity metric discriminatively, with application to face verification. In *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, volume 1, pages 539–546, 2005. doi: 10.1109/CVPR.2005.202. URL <https://ieeexplore.ieee.org/document/1467314>.
- Mariona Coll Ardanuy and Caroline Sporleder. Toponym disambiguation in historical documents using semantic and geographic features. In *Proceedings of*

the 2nd International Conference on Digital Access to Textual Cultural Heritage (DATECH), pages 175–180, 2017. doi: 10.1145/3078081.3078096.

Maud Ehrmann, Matteo Romanello, Alex Flückiger, and Simon Clematide. Extended overview of CLEF HIPE 2020: Named Entity Processing on Historical Newspapers. In *Proceedings of the CLEF 2020 Conference and Labs of the Evaluation Forum*, 2020. URL https://ceur-ws.org/Vol-2696/paper_253.pdf.

Maud Ehrmann, Matteo Romanello, Sven Najem-Meyer, Antoine Doucet, and Simon Clematide. Extended overview of HIPE-2022: Named Entity Recognition and Linking in Multilingual Historical Documents. In *Proceedings of the CLEF 2022 Conference and Labs of the Evaluation Forum*, 2022. URL <https://ceur-ws.org/Vol-3180/paper-83.pdf>.

Stephen James Gadd and Alexis Litvine. Index villaris, 1680, 2024. URL <https://doi.org/10.5281/zenodo.10659931>.

Zhouhan Lin, Minwei Feng, Cicero Nogueira Santos, Mo Yu, Bing Xiang, Bowen Zhou, and Yoshua Bengio. A structured self-attentive sentence embedding. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017. URL <https://arxiv.org/abs/1703.03130>.

David R. Mortensen, Patrick Littell, Akash Bharadwaj, Kartik Goyal, Chris Dyer, and Lori Levin. Panphon: A resource for mapping IPA segments to articulatory feature vectors. In *Proceedings of the 26th International Conference on Computational Linguistics (COLING)*, pages 3475–3484, 2016. URL <https://aclanthology.org/C16-1328/>.

David R. Mortensen, Siddharth Dalmia, and Patrick Littell. Epitran: Precision G2P for many languages. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC)*, pages 2634–2639, 2018. URL <https://aclanthology.org/L18-1140/>.

Jonas Mueller and Aditya Thyagarajan. Siamese recurrent architectures for learning sentence similarity. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*, pages 2786–2792, 2016. URL <https://dl.acm.org/doi/10.5555/3016100.3016291>.

Paul Neculoiu, Maarten Versteegh, and Mihai Rotaru. Learning text similarity with siamese recurrent networks. In *Proceedings of the 1st Workshop on Representation Learning for NLP*, pages 148–157, 2016. URL <https://aclanthology.org/W16-1617/>.

- Lawrence Philips. Hanging on the metaphone. *Computer Language*, 7(12):39–43, 1990.
- Lawrence Philips. The double metaphone search algorithm. *C/C++ Users Journal*, 18(6):38–43, 2000. doi: 10.5555/349124.349132.
- Qinjun Qiu, Haiyan Li, Xinxin Hu, Miao Tian, Kai Ma, and Yunqiang Zhu. A deep neural network model for chinese toponym matching with geographic pre-training model. *International Journal of Digital Earth*, 17(1):2353111, 2024. doi: 10.1080/17538947.2024.2353111. Proposal of the GeoBERT model.
- Taraka Rama. Siamese convolutional networks for cognate identification. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 1018–1027, Osaka, Japan, 2016. The COLING 2016 Organizing Committee. URL <https://aclanthology.org/C16-1097/>.
- Gabriel Recchia and Max Louwerse. A comparison of string similarity measures for toponym matching. In *Vagueness in Communication*, pages 54–67. Springer, 2013. doi: 10.1007/978-3-642-18446-8_4.
- Robert C. Russell. Index. U.S. Patent 1,261,167, 1918. Patented Apr. 2, 1918.
- Tomer Sagi, Moran Zaga, Sinai Rusinek, Marcell Richard Fekete, Johannes Bjerva, and Katja Hose. Utilizing phonetic similarity for cross-source and cross-language toponym matching: a benchmark and prototype. *Language Resources and Evaluation*, 59(3):2427–2451, 2025. doi: 10.1007/s10579-025-09812-9. Associated with the MEHDIE project.
- Rodrigo Santos, Patricia Murrieta-Flores, Pablo Calafiore, and Bruno Martins. Learning to combine multiple string similarity metrics for effective toponym matching. *International Journal of Digital Earth*, 11(9):949–965, 2018. doi: 10.1080/17538947.2017.1371253.
- Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823, 2015. doi: 10.1109/CVPR.2015.7298682. URL <https://api.semanticscholar.org/CorpusID:206592766>.
- Mike Schuster and Kuldip K. Paliwal. Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11):2673–2681, 1997. doi: 10.1109/78.650093.

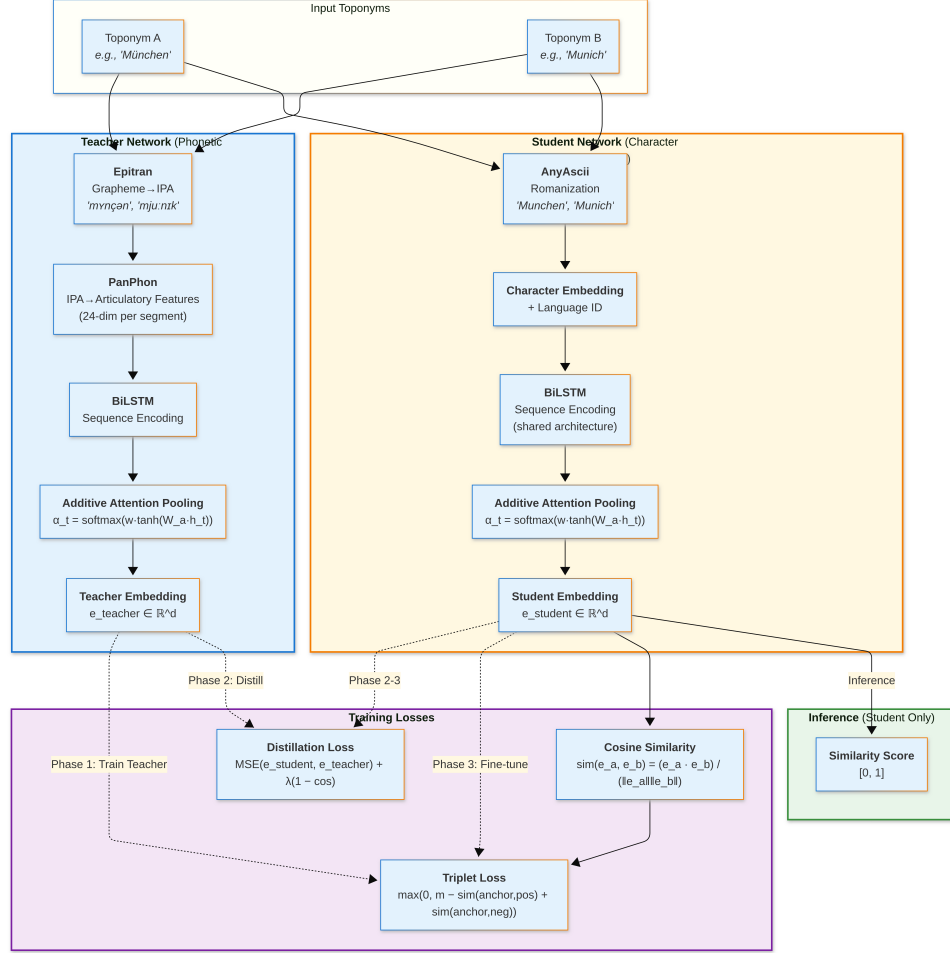


Figure 1: Student-Teacher architecture for phonetic similarity learning. The Teacher network processes IPA articulatory features via Epitrans and PanPhon; the Student network processes romanized characters with language embeddings. Both use BiLSTM encoding followed by additive attention pooling. During training, the Student learns to approximate Teacher embeddings through knowledge distillation (Phases 1–3). At inference, only the Student network is required, outputting cosine similarity scores.

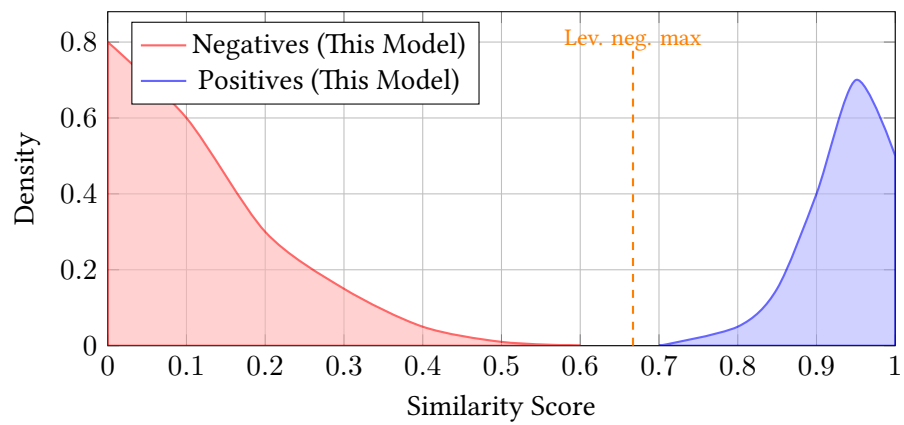


Figure 2: Similarity score distributions for positive and negative pairs. The trained model concentrates negative pairs near zero (95th percentile = 0.485), creating a wide margin for threshold selection. Levenshtein negatives extend to 0.667, reducing the viable operating range.